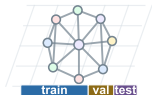


HARP-Select: Validation-Calibrated Routing for Complementary Residual Graph Experts

Shared graph split
fixed once for both experts



MATCHED PROTOCOL

- ⊖ fixed train/val/test masks
- ⊖ identical optimizer
- ⊖ identical training budget

shared (G, X) and labels

Matched-budget residual experts

parallel training; no cross-branch information

VIEW FILTER BANK FUSION VAL. SCORE

HARP-GNN



a_H validation

HARP-ESep



a_E validation

COMPLEMENTARY RESIDUAL BASES



H: residual node fields

$$H_k = \hat{A}^{k-1} X - \hat{A}^k X$$



E: ego-separated fusion

$$B_k = \tilde{A}^k B_0, [B_0, U_i]$$

Validation route and test

select first, evaluate second

a_H

a_E

$$\Delta_{val} = a_E - a_H$$

retain H

select E

switch only if $\Delta_{val} > 1.96 SE$

(H)

(E)

e^*

test labels revealed after branch lock



f_{e^*}

$\hat{y}$

y_{test}

score($\hat{y}, y$)